Chi-Sheng (Michael) Chen (陳麒升)

Research Engineer | LLM Agents & Evaluation | Time-Series & Multimodal ML | Foundation Models
+1 (351) 244-1084 | m50816m50816@gmail.com | LinkedIn | GitHub | Google Scholar
Based in Cambridge, MA, USA

Research engineer building LLM-agent systems and time-series/multimodal models, with an emphasis on verifiable evaluation. Building an ASR → LLM-agent → prediction pipeline over 4M+ EMS records at Harvard/BIDMC; 3rd of 1,300+ teams at Berkeley RDI AgentX-AgentBeats 2026; led LLM token-efficiency research for a quant-trading firm; EEG treatment-response model in routine clinical use. 49 publications, 515 citations; ICML Gold Reviewer.

Experience

Harvard Medical School & Beth Israel Deaconess Medical Center — MA, USA
Research Affiliate (primary appointment), Advisor: Dr. Gabriel Brat — *Nov 2024 – Present*

- Building a real-time EMS trauma-triage pipeline (ASR → LLM-agent event extraction → structured prediction) over **4M+ patient records** using multimodal AI.
- Co-developed webcam-based open-surgical skill assessment (hand keypoint detection; CLINICCAI 2026 poster) and a capture-clock methodology for browser webcam gaze (DEMI 2026 spotlight talk).

WhaleForce (subsidiary of VICI Holdings) — Taipei, Taiwan
Fractional Token Efficiency Research Lead (Advisor) — *Aug 2026 – Present*

- Advising on LLM token-efficiency research and multi-agent architecture for AI-driven quantitative trading.

Omnis Labs — Remote
Co-Founder — *Sep 2024 – May 2026*

- AI-driven DeFi liquidity protocol: time-series models drive automated-market-maker (AMM) liquidity positioning, with a backtesting system; ran in production on WBTC/USDC and ETH/USDC pools at 50%–120% base fee APR.

Neuro Industry, Inc. — CA, USA
Researcher, Co-Founder & CTO — *Mar 2024 – Jan 2025*

- Built GCP-based MLOps pipeline, enabling training on 10K+ EEG samples.
- 3 research papers: quantum machine learning for EEG signal processing (2) and EEG-to-image diffusion (1).

Department of Computer Science, National Yang Ming Chiao Tung University — Hsinchu, Taiwan
Research Assistant, Advisor: Prof. Chun-Shu Wei — *Dec 2023 – Aug 2024*

- Developed MUSE, a multimodal contrastive learning framework for EEG-based image recognition (34 citations).
- Research on nonstationary EEG dynamics and foundation models for brain-computer interfaces.

MediaTek Inc. — Hsinchu, Taiwan
Digital IC R&D Engineer — *Dec 2021 – Oct 2023*

- Microprocessor IP for flagship 5G smartphone display SoCs: RTL design of a hardware-virtualization architecture and IP verification with UVM/SystemVerilog.

Department of Psychiatry, Taipei Veterans General Hospital — Taipei, Taiwan
Research Assistant, Advisor: Prof. Cheng-Ta Li, MD — *Sep 2019 – Jun 2021*

- Built ML models on clinical EEG from major-depressive-disorder patients to predict treatment response; the model (M.S. thesis work) is in routine outpatient use at Taipei VGH's Precision Depression Intervention Center.

Other: OpenAI (contractor via Mercor) — AI Trainer, RLHF preference data & model-behavior evaluation — *Mar 2025 – Oct 2025*

Earlier: NTU Hospital Surgery (surgical-video detection); Max Planck Institute CPfS, Dresden (intern) — *2018 – 2021*

Agent & LLM Engineering

Berkeley RDI AgentX-AgentBeats — 3rd Place / 1,300+ teams (2026). *MadGAA*, a multi-agent OSCE medical evaluation system: role-specialized agents (examiner, patient, judge) + verifiable scoring rubric (RAG + tool use + verifier). [code] [leaderboard]
US-stock LLM agent suite (deployed). Browser agent as an explicit PLAN→LOCATE→ACT→VERIFY state machine with fault-injection recovery; layered SEC 10-K extractor; 22 research agents + 11 placebo controls with lookahead-free backtests, per-provider cost ledger, and eval dashboards.
Robotics Workcell Optimizer (live). NL → spec → robot selection → 2D layout → 3D preview via multi-LLM abstraction (Claude/OpenAI/Gemini, validation-repair loop) + OR-Tools CP-SAT; ISO 13855 safety as inviolable hard constraints. [code] [demo]
Also: *flyhypo* (connectome + PubMed + LLM → grounded, falsifiable neuron hypotheses) [code]; *paper-evidence* (verbatim-quote gate + cross-family judge to block LLM hallucination) [code]; two live NYCU RAG physics tutors (Next.js + Gemini + pgvector) [live].

Education

National Taiwan University (NTU) — Taipei, Taiwan
Master of Science in Biomedical Electronics and Bioinformatics — *Sep 2019 – Jun 2021*

National Chiao Tung University (NCTU) — Hsinchu, Taiwan
Bachelor of Engineering in Electrophysics — *Sep 2015 – Jun 2019*
Bachelor of Science in Interdisciplinary Science Degree Program

Skills

LLM & Agents: LangChain/LangGraph, ReAct & function calling, multi-agent orchestration, RAG (pgvector/FAISS/Qdrant), OpenAI/Anthropic/Gemini SDKs, OR-Tools
Training & Serving: PyTorch, HuggingFace, LoRA/QLoRA, SFT/DPO/RLHF, vLLM, MLflow, scikit-learn
Engineering: Python, C/C++, JavaScript, Rust, SQL, Git, Docker, GCP, AWS
Languages: Chinese (Native), English (Fluent)

Selected Publications

(49 total, 515 cites, h-index 13 — full list on Google Scholar)

[First author unless noted]

Chen, C.S., et al. "FreqLens: Interpretable Frequency Attribution for Time Series Forecasting" *Under review*, ICLR 2027; arXiv:2602.08768.
Chen, C.S., et al. "Large Cognition Model: Towards Pretrained EEG Foundation Model" arXiv preprint (2025), arXiv:2502.17464 — **30 citations**.
Chen, C.S., Wei, C.S. "Mind's Eye: Image Recognition by EEG via Multimodal Similarity-Keeping Contrastive Learning" arXiv preprint (2024), arXiv:2406.16910 — **34 citations**.
Chen, C.S., et al. "Quantum Reinforcement Learning-Guided Diffusion Model for Image Synthesis via Hybrid Quantum-Classical Generative Model Architectures" IEEE ICASSP 2026, **oral**.
Chen, C.S., Brat, G.A. "Measuring Browser Webcam Gaze Honestly: A Capture-Clock Methodology and Open Reference Implementation" MICCAI DEMI Workshop 2026, **spotlight talk**; arXiv:2608.11566.
Li, C.T., **Chen, C.S., et al. "Prediction of antidepressant responses to non-invasive brain stimulation using frontal electroencephalogram signals: Cross-dataset comparisons and validation"** Journal of Affective Disorders (2023). [2nd author; model deployed at Taipei VGH PreDIC]

Recognition & Service

Awards: NVIDIA GTC 2026 Top 8 Finalist (GPU-accelerated Chinese brain-to-text interface on H100); Berkeley RDI AgentX-AgentBeats 2026, 3rd of 1,300+ teams; 20th National Innovation Award (Taiwan, 2023), clinical research category.
Talks: IEEE ICASSP orals (two in 2026, one in 2025); MICCAI 2026 DEMI spotlight talk and CLINICCAI poster; NVIDIA GTC 2026 (two posters); guest lectures at NTU CSIE, NYCU ECE and Chang Gung University BME (2026).
Reviewing: NeurIPS, ICML (Gold Reviewer), AAAI, KDD, MICCAI, ICASSP; 44 journals incl. IEEE TPAMI, IEEE TNNLS and Neural Networks.